MATH SURVEY

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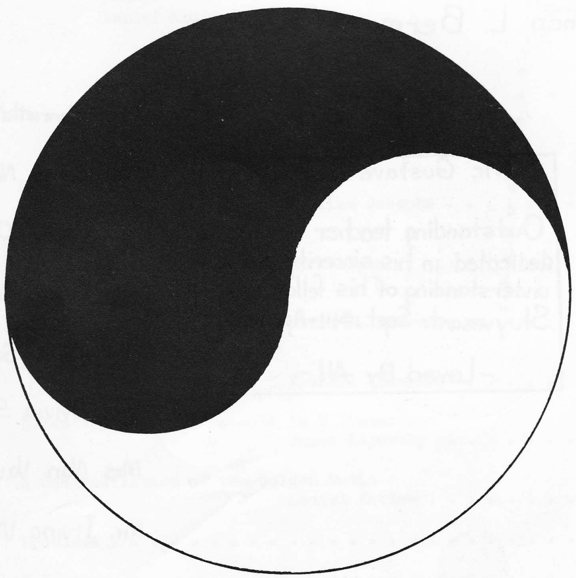

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THE MATH SURVEY

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Dedication

To the Teachers of the Department of Mathematics who have encouraged our interest in the Queen of Sciences.

Pythagorean Triples

William Josephs

It is well known that the numbers x, y, z given by the formulas

$$x = u^2 - v^2$$
$$y = 2uv$$
$$z = u^2 + v^2$$

where u and v are positive integers and $u > v$, satisfy the Pythagorean formula

$$x^2 + y^2 = z^2.$$

Hence, x and y may be considered as the legs of a right triangle with integral sides and hypotenuse z. To investigate the properties of all such Pythagorean number triples (x, y, z), I, as a member of the Columbia University Science Honors Program, programmed the IBM 650 computer to solve these equations for consecutive values of u and v. I then analyzed the results in connection with the following theorems and with the help of the following definitions and notation:

1. primitive triangle - a triangle (x, y, z) where the sides have no common factor.
2. family of triangles - a group of triangles of the form (nx, ny, nz) where (x, y, z) is a primitive triple (triangle) and n is any positive integer.
3. coefficient of similarity - the greatest common factor of the sides of a triangle. If each side of a triangle is divided by the coefficient of similarity, the resulting numbers form a primitive triangle.

The letters u and v, primed or unprimed, will always represent solutions of the above equations for some triangle. For example, "There exist u, v for a triangle such that . . ." will mean "There exist numbers u, v that satisfy the above equations for a triangle such that . . ."

THEOREM 1. In a primitive triangle (x, y, z), no two sides have a common factor.

proof: Suppose x and y have the common factor d. Then $x = ad$, $y = bd$,

$$x^2 + y^2 = z^2 = d^2(a^2+b^2).$$

Thus, we see that d^2 divides z^2, by which we deduce that d divides x, y, and z, and therefore the triangle is not primitive. The reasoning is similar if x, z or y, z have a common factor.

THEOREM 2. In a primitive right triangle (x, y, z), one and only one of the legs x and y are odd.

proof: By Theorem 1, x and y cannot both be even. Furthermore, they cannot both be odd. For, if x and y are odd, we may set $x = 2m+1$, $y = 2n+1$, whence

$$z^2 = x^2 + y^2$$
$$= 4(m^2+n^2+m+n)+2.$$

Hence, z^2 is even but is not divisible by four. But this is a contradiction, since all even squares are divisible by four. Therefore, one of the legs is even and one is odd. Henceforth, x will denote the odd leg.

THEOREM 3. For every primitive triple (x, y, z), there exists one and only one u, v.

proof: From $x^2+y^2=z^2$ we obtain

$$y^2=z^2-x^2=(z+x)(z-x)$$

$$\left(\tfrac{1}{2}y\right)^2=\tfrac{1}{2}(z+x)\tfrac{1}{2}(z-x)$$

Since z and x are both odd, $\frac{1}{2}(z+x)$ and $\frac{1}{2}(z-x)$ are both integers. Furthermore, they have no common factor, for if they did, that factor would divide both their sum $\frac{1}{2}(z+x)+\frac{1}{2}(z-x)=z$ and their difference $\frac{1}{2}(z+x)-\frac{1}{2}(z-x)=x$ which is impossible by Theorem 1. Now, if two numbers a and b have no common factors and their product ab is a perfect square, then each must be a perfect square. For each prime factor of a appears an even number of times in ab but does not appear in b. Hence, a contains only an even powers of primes, and is therefore a square and similarly for b. Therefore, $\frac{1}{2}(z+x)$ and $\frac{1}{2}(z-x)$ are perfect squares. Let us write

$$\tfrac{1}{2}(z+x)=u^2, \quad \tfrac{1}{2}(z-x)=v^2.$$

Then we may solve for x, y, z to obtain the desired equations

$$\begin{aligned} x &= u^2-v^2 \\ y &= 2uv \\ z &= u^2+v^2. \end{aligned}$$

Suppose there is another solution u', v'. Then

$$\begin{aligned} u'^2-v'^2 &= u^2-v^2 \\ u'^2+v'^2 &= u^2+v^2 \\ 2u'^2 &= 2u^2 \\ 2v'^2 &= 2v^2 \\ u' &= u \\ v' &= v. \end{aligned}$$

Hence, there is one and only one solution u, v. u and v can have no common factors, and must be of opposite parity since $x=u^2-v^2$ is odd.

THEOREM 4. For a given triangle, u and v exist if and only if the coefficient of similarity is a perfect square.

proof: Let (nx, ny, nz) be the given triangle where (x, y, z) is primitive. By Theorem 3, u', v' exist for (x, y, z). If u, v exist for (nx, ny, nz), we must have

$$\begin{aligned} u^2-v^2 &= nu'^2-nv'^2 \\ u^2+v^2 &= nu'^2+nv'^2 \\ 2u^2 &= 2nu'^2 \end{aligned}$$

whence n is a perfect square. Conversely, if n is a perfect square, we may write

$$\tfrac{1}{2}(z+x)n=u^2, \quad \tfrac{1}{2}(z-x)n=v^2$$

to obtain the desired solution. Hence, the theorem is proved.

THEOREM 5. If u, v exist for (x, y, z), $u-v=1$ if and only if $z-y=1$.

proof: $y=2uv=2v(v+1)=2v^2+2v$ if $u-v=1$. Also,

$z=u^2+v^2=(v+1)^2+v^2=2v^2+2v+1.$

Hence, $z-y=1$. Conversely, if $z-y=1$, we have

$$\begin{aligned} (u^2+v^2)-2uv &= 1 \\ (u-v)^2 &= 1 \\ u-v &= \pm 1, \end{aligned}$$

and, since $u>v$, $u-v=1$.

Symbolic Logic

Mark Alper

In mathematics one often encounters data from which a conclusion must be drawn. This is frequently a relatively simple operation. If you were given (a) no experienced person is incompetent, (b) Jenkins always blunders, and (c) no competent person blunders, it would not be hard to deduce that Jenkins must be inexperienced. But if you were given nine ideas and nine statements combining them in different ways, and had to draw a conclusion from them, problems might arise. A mathematical method of attacking such problems is symbolic logic.

Since logic itself is being developed as a "deductive theory," we must begin with undefined terms and axioms and then derive the defined terms and theorems. To facilitate writing, we shall assign a symbol to each term. Also, as with most deductive theories, there are several different axiomatic systems and sets of undefined terms which determine the same theory. An undefined term in one system may be defined in another, and an axiom in one may be a theorem in another.

The first undefined term is the proposition, or statement, such as "Two is an even prime." These are denoted by letters of the alphabet: True, T; False, F; and, $\wedge$; or, $\vee$; implies, $\rightarrow$; equivalent, $\leftrightarrow$; and not, $\sim$, are the seven other terms which we shall leave undefined, although three of them may actually be defined in terms of the others. The last five terms are called connectives.

The union of two propositions by "and" is called the conjunction of the propositions, while two propositions conjoined by "or" constitute a disjunction. The implication, equivalence and negation of propositions are similarly defined in terms of the respective undefined terms. It is important to note that two propositions do not have to be related in order to be combined, as they do in English usage. Thus, "If the sun is shining, 3 x 6 = 18," is an acceptable implication.

An important part of symbolic logic is the calculus of propositions, dealing with the truth-values and relationships of propositional functions. A propositional function is a combination of terms including connectives and propositional variables which would become a proposition if propositions were substituted for the variables. Thus, for example, "A implies B" is a propositional function if "A" and "B" are variables. To decide whether a case of the function is true we must examine the truth-values of the constituent propositions.

For a propositional function of two variables, there are four possible combinations of truth-values, since each may be true or false. With three variables, there are eight combinations, and with n there are 2^n combinations. A systematic, though tedious, method of dealing with propositional functions was devised by the American logician, Peirce. This is the method of truth tables, in which all the combinations and the corresponding values of the function are written out.

We may now illustrate the use of symbolism in the problem mentioned in the first paragraph.

We begin by assigning letters to the separate statements so that the hypothesis may be rewritten in symbolic form. Let e be "a person is experienced," J be "a person named Jenkins," c be "a person is competent," and b be "a person is a blunderer."

Then the statements become A. $e \rightarrow c$, B. $J \rightarrow b$, C. $c \rightarrow \sim b$. We wish to prove

$$(e \rightarrow c) \wedge (J \rightarrow b) \wedge (c \rightarrow \sim b) \rightarrow (J \rightarrow \sim e).$$

continued on page 9.

Bernstein's Theorem

David S. Lawrence

One of the fundamental theorems in the theory of transfinite numbers states that two sets are equivalent if each is equivalent to a subset of the other. This theorem, asserted by Cantor, was first proved by Bernstein. The proof usually given is based on the following lemma: '' If a set S is equivalent to a subset T, then a subset of S which includes T is also equivalent to them.'' In this paper I shall present a more direct proof, but we must first establish the elements of the theory.

Two sets are called equivalent if and only if there is a one-one transformation from one to the other. We shall denote this transformation by $f(X) = Y$, where X and Y are the sets, $X \sim Y$. Consequently, if $X_i \sim Y_i (i = 1,2,3,\ldots)$ then $\bigcup^{\infty} X_i \sim \bigcup^{\infty} Y_i$, where the notation $\bigcup^{\infty} X_i$ signifies a set including the elements of all X_i, if they are disjoint.

We are given sets A and B, with $f(A) = B_1$, $g(B) = A_1$, and B_1 and A_1 are subsets of B and A respectively. From these we form a sequence of subsets A_n and B_n according to:

1) $fg(B_n) = f(A_{n+1}) = B_{n+2}$

2) $f(A_n - A_{n+1}) = B_{n+1} - B_{n+2}$

3) $gf(A_n) = g(B_{n+1}) = A_{n+2}$

4) $g(B_n - B_{n+1}) = A_{n+1} - A_{n+2}$. $\qquad A = A_0,\ B = B_0,\ n = 0,1,2,\ldots$

Finally we form a new sequence of sets A_n^*, and a sequence B_n^*: $A_n^* = A_n - A_{n+1}$, and $B_n^* = B_n - B_{n+1}$. If $\prod_0^\infty A_n$ denotes the set of elements which are members of all the A_n, then it is clear that $A = (\prod_0^\infty A_n) \cup (\bigcup_0^\infty A_n^*)$, and similarly for B. By (2) and (4) $A_n^* \sim B_{n+1}^*$, and $B_n^* \sim A_{n+1}^*$,
from which we easily derive $\bigcup_0^\infty A_n^* \sim \bigcup_0^\infty B_n^*$, since the B_n^* and the A_n^* are all disjoint. By (1) and (3) we see that any element a of all the A_n is transformed onto an element b of all the B_n, that is, $\prod_0^\infty A_n \sim \prod_0^\infty B_n$. The conclusion of the theorem is an immediate consequence of the last two results above. $A \sim B$.

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Homogeneous Coordinates

Michael Ackerman

Although the usual Cartesian coordinates satisfy the needs of elementary geometry, when we wish to consider projective geometry these will not suffice and homogeneous coordinates are found convenient.

For finite points, the new coordinates will be defined in terms of the Cartesian coordinates.

DEFINITION 1. Homogeneous coordinates of the finite point (x,y) are any three numbers, x_1, x_2, x_3, for which, $x_1/x_3 = x$, $x_2/x_3 = y$. Thus, for any $r \neq 0$, (rx, ry, r) are homogeneous coordinates of (x, y).

A finite point has, then, infinitely many sets of homogeneous coordinates. Each two sets are proportional, and the third coordinate of every set is different from zero. Conversely, any three numbers, the third of which is not zero, are homogeneous coordinates of a definite finite point, namely the point (x, y), where $x = x_1/x_3$, $y = x_2/x_3$.

In geometry, parallel lines might be defined as lines for which interior angles on the same side of a transversal are supplementary. An ideal point might then be defined as the point of intersection of two parallel lines.

DEFINITION 2. Corresponding to each set of parallel lines there is an ideal point, or point at infinity. This point lies on each line of the set and on no other finite line. It is known as the point at infinity in the given direction.

Inasmuch as there are infinitely many directions in the plane, there are infinitely many ideal points. It is natural to think of these points as constituting a curve. But what type of curve?

DEFINITION 3. The ideal points of the plane lie on an ideal straight line, the line at infinity in the plane.

These considerations remove certain exceptions that previously appeared in our geometry. For example, we may now say that just as every two points determine exactly one line, two lines determine exactly one point.

We found that to the finite points there correspond as coordinates all the number triples (x_1, x_2, x_3) for which $x_3 \neq 0$. There remain the triples $(x_1, x_2, 0)$. We shall show that these are suitable coordinates for the points at infinity.

Consider the point at infinity in the direction whose slope is m. An arbitrary line through this point, that is, an arbitrary line of slope m, has the equation

$$y = mx + b.$$

Let the variable point $P(x,y)$ recede indefinitely in either direction. If, then, homogeneous coordinates of P can be found which have limits which are independent of b, that is, independent of the particular line of slope m chosen, these limits will be reasonable coordinates for the point at infinity.

Cartesian coordinates of P are $(x, mx + b)$. A set of homogeneous coordinates of P is therefore $(x, mx + b, 1)$. Here the first two coordinates become infinite with x. However, the coordinates $(1, m + b/x, 1/x)$ approach $(1, m, 0)$ independently of b.

Accordingly, the coordinates of the ideal point in the direction of slope m are $(r, rm, 0)$, where r is any nonzero number.

The preceding discussion does not include the ideal point on the y-axis, but the reader will see that as coordinates of points on the y-axis are of the form

$$(0, y, 1) = (0, r, r/y), \quad r \neq 0,$$

and since r/y approaches zero as y becomes infinite, the coordinates of the ideal point are $(0, r, 0)$.

We have now assigned to all the points of the extended plane homogeneous coordinates, and have thereby exhausted all triples, except $(0, 0, 0)$, which has no point.

Now consider the equation of a straight line:

$$a_1 x + a_2 y + a_3 = 0.$$

If we substitute in this equation the values of x and y in terms of the homogeneous coordinates we obtain for the equation of a straight line:

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0.$$

Note that a necessary and sufficient condition that (x_1, x_2, x_3) be an ideal point is that

$$x_3 = 0.$$

The fact that 3) is a special case of 2) provides justification for Definition 3.

As a consequence of the above considerations (a_1, a_2, a_3) are homogeneous coordinates of a straight line. The duality between point and line in projective geometry is thus preserved in this coordinate system.

There are many uses of homogeneous coordinates, and they simplify many of the formulas of ordinary analytic geometry, but space does not permit a discussion of this topic.

Symbolic Logic - continued from page 6

We might set up a truth table to show this, but it would be impractical because of the nine columns and sixteen rows needed. Instead, we may derive theorems which eliminate the use of tables.

As in other systems, these theorems must hold for all cases. The theorems we will use are

$$(p \to q) \wedge (q \to r) \to (p \to r)$$

$$(a \to b) \to (\sim b \to \sim a)$$

The first is the Law of Syllogism, and the second is the contrapositive.

Truth Table for contrapositive

a	b	(a → b)	→	(~ b	→	~ a)
T	T	T	T	F	T	F
T	F	F	T	T	F	F
F	T	T	T	F	T	T
F	F	T	T	T	T	T

Then here is the proof of the problem.

1. $e \to c,\ J \to b,\ c \to \sim b$ — **1. Given**
2. $b \to \sim c,\ \sim c \to \sim e$ — **2. Contrapositive of C and A**
3. $\therefore b \to \sim e$ — **3. Syllogism**
   $\therefore J \to \sim e$

Euler's Polyhedral Formula in N Dimensions

James Lepowsky

Before generalizing Euler's polyhedral formula, we must clarify the concept of n-dimensional space. Analytically, we may define a point in a space of n dimensions as an ordered set of n real numbers. The solution of a set of n simultaneous linear equations in n unknowns may be considered the coordinates of the point of intersection in n-space of n hyperplanes (spaces of (n-1) dimensions), each the geometric locus of one equation.

As a second method of developing this theory, we may generalize Hilbert's axioms of three-dimensional space. One of these states that two planes with one point in common have a second point in common. If this is replaced by an axiom postulatiog the existence of at least one point not in the region determined by the other axioms, a four-dimensional space is determined. Further extension may be used to define space of n dimensions.

A polytope in n dimensions, the analogue of a polygon in two dimensions, and a polyhedron in three, is bounded by portions of hyperplanes which intersect in boundaries of (n-2) dimensions. Only simple polytopes will be considered here. These have properties analogous to those of the Eulerian polyhedrons -polyhedrons without ring-shaped faces, (fig. 1); rings, (fig. 2); cavities (fig. 3).

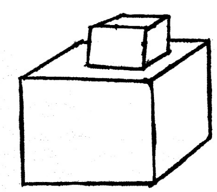


Figure 1.

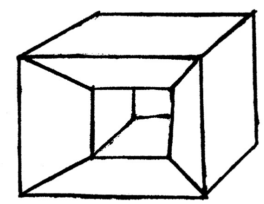


Figure 2.

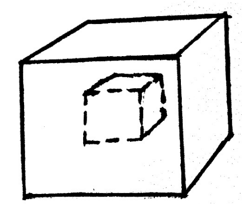


Figure 3.

Simple n-polytopes may be topologically deformed (stretched) onto the surface of an n-dimensional sphere, and so on for all boundary figures of all dimensions. This condition excludes polytopes that are not simple.

Consider Euler's polyhedral formula, true in general for only simple (Eulerian) polyhedrons,

$$V - E + F = 2$$

where V represents the number of vertices; E, the number of edges; and F, the number of faces. The analogue of this formula for polygons states that

$$V - E = 0$$

since the number of vertices of a polygon equals the number of edges. For the one-dimensional case, a line segment, we have

$$V = 2$$

These formulas may be expressed as follows:

For n = 3 $\quad P_0 - P_1 + P_2 - P_3 = 1$

For n = 2 $\quad P_0 - P_1 + P_2 = 1$

For n = 1 $\quad P_0 - P_1 = 1$

where P_i is the number of boundary figures of i dimensions; $P_n = 1$ for a polytope of n dimensions.

We may conjecture that for a simple polytope of n dimensions,

$$\sum_{i=0}^{n} (-1)^i P_i = 1$$

where P_i is defined as in the preceding paragraph. It will be proved by mathematical induction that this formula does in fact hold for all n.

By the axiom of induction, if the integer 1 is in a given class of natural numbers, and if for every integer k in the class, k+1 is in the class, then the class consists of all the natural numbers. We must therefore show that the generalization formula above holds for n = 1, and that if it holds for n = k, then it holds for n = k+1.

When n = 1, the formula states that

$$P_o - P_1 = 1$$

for polytopes of one dimension (line segments). This is true, since $P_o = 2$ and $P_1 = 1$.

We must show now that if

$$\sum_{i=0}^{k} (-1)^i P_i = 1 \qquad (1)$$

for every simple k-polytope, then

$$\sum_{o}^{k+1} (-1)^i P_i = 1 \qquad (2)$$

for any simple (k+1) polytope.

We first note that it is possible to remove any hyperface of a simple (k+1) polytope to obtain a figure called a network of k dimensions. We must show that for this figure,

$$\sum_{o}^{k} (-1)^i P_i = 1$$

since $P_{k+1} = 1$ and P_k and P_{k+1} have opposite signs in equation (2).

Consider a simple k-polytope with a_o vertices, a_1 edges, a_2 faces,..., a_{k-1} (k-1)-polytopes and a_k k-polytopes ($a_k = 1$). We have assumed that

$$\sum_{o}^{k} (-1)^i a_i = 1$$

by (1). By constructing k-polytopes on the hyperfaces of this figure, we shall be able to build any given network of k dimensions, for which we must prove that

$$\sum_{o}^{k} (-1)^i T_i = 1$$

where T_i is the total number of i-polytopes added.

Let us construct a kpolytope with b_o vertices, b1 edges,......, and b_k k-polytopes $b_k = 1$ so that this and the original polytope have c_{k-1} hyperfaces in common ($c_{k-1} > 0$).

In the network of (k-1) dimensions commnn to both, there are c_o vertices, c_1 edges...., and c_{k-1} (k-1)-polytopes. Thus the number of vertices added is $b_o - c_o$; the number of edges added, $b_1 - c_1$;...; the number of (k-1)-polytopes added, $b_{k-1} - c_{k-1}$ and the number 1
of k-polytopes added, b_k or 1. We must show that

$$\sum_{o}^{k} (-1)^i (b_i - c_i) = 0$$

or that

$$\sum_{o}^{k} (-1)^i b_i = \sum_{o}^{k-1} (-1)^i c_i \qquad (3)$$

But the left member of (3) is 1 by assumption (1), and it remains to show only that the right member equals 1.

We must therefore prove that if

$$\sum_{o}^{k} (-1)^i P_i = 1 \qquad (4)$$

for every simple k-polytope, then

$$\sum_{o}^{k-1} (-1)^i P_i = 1$$

for any network of (k-1) dimensions. This is true since $P_k = 1$, and since P_k and P_{k-1} have opposite signs in (4). As before, we can conclude that the removal of a boundary (k-1)-polytope from some k-polytope leads to the desired result for any given network.

In order to complete the proof by induction, we note that the repeated construction of k-polytopes on the original one does not change the value of

$$\sum_{o}^{k} (-1)^i P_i$$

This expression remains equal to 1 for the entire network. As noted before, a (k+1)-polytope is formed from this network by the addition of the last hyperface, and

$$\sum_{o}^{k+1} (-1)^i P_i$$

(The methods of addition of the simple k-polytopes and completion of the figure insure that the resulting (k+1)-polytope is simple.) Q. E. D.

The above generalization of Euler's formula may be applied in advanced topology. An elementary application is a generalization of the proof given in [1], page 240, of the theorem that there are no more than five regular polyhedrons.

[1] Courant, Richard & Robbins, Herbert, What is Mathematics? New York: Oxford University Press, 1941.

[2] Manning, Henry Parker, Geometry of Four Dimensions. New York: Dover Publications, 1956.

[3] Sommerville, D. M. Y., An Introduction to the Geometry of N Dimensions. New York: Dover Publications, 1958.

A Generalization of the Golden Ratio

Daniel Kotlow

The positive root of the equation

$$x^2 = x + 1.$$

has been the object of much study and wonder since ancient times. It occurs in many varied geometrical and arithmetical situations in the form of a ratio, and therefore has been called the "golden ratio." The purpose of this article is to define and investigate a generalization of the golden ratio, to which we would like to attribute properties which are generalizations of those of the golden ratio. We procede by defining a number $T(n)$ through a generalization of the above equation.

DEFINITION 1. $T(n)$ is the positive root of the equation

$$x^n = \sum_{i=0}^{n-1} x^i. \qquad (1)$$

$T(2)$ is obviously the golden ratio. Then, we have the basic theorem regarding $T(n)$.

THEOREM 1. For each positive integer n, $T(n)$ exists and is unique.

proof: By Descartes' Rule of Signs, equation (1) has at most one positive root. Therefore, it has either zero or one root. If n is even, the product of the roots is -1. Therefore, the number of negative roots is odd. Since the number of imaginary roots is even and there are n roots, the number of positive roots is odd. If n is odd, the product of the roots is $+1$. Hence, there is an even number of negative roots. Since the number of imaginary roots is even and there are n roots, there must be an odd number of positive roots. Hence, in all cases there is exactly one positive root, q.e.d.

THEOREM 2. $1 \le T(n) < 2 \qquad (2)$

proof: If $x<1$ in (1), each term on the right side is greater than the left side, and the equality cannot hold. Observing that the right side of (1) is a geometric progression, we write

$$x^n = \frac{x^n - 1}{x - 1} \qquad (1')$$

$$x^{n+1} - 2x^n = -1 \qquad (1'')$$

and

$$x^n(x-2) = -1 < 0$$

If $x = T(n)$, $x > 0$ by definition, and we must have $x<2$. Hence, (2) is established.

It is now a simple task to compute values of $T(n)$ by Horner's method. The first five values are $T(1) = 1.000$, $T(2) = 1.618$, $T(3) = 1.837$, $T(4) = 1.928$, $T(5) = 1.937$. These figures suggest some further theorems, but before stating them we will need a definition.

DEFINITION 2. $T^*(n)$, for positive n, is the least positive root other than unity of

$$x^{n+1} - 2x^n + 1 = 0. \qquad (3)$$

Since (3) is the same as (1''), it is clear that $T(n) = T^*(n)$ for integral n, for then there cannot be more th an two positive roots.

LEMMA. $T^*(n) > \frac{2n}{n+1} \qquad (4)$

proof: Let $f(x) = x^{n+1} - 2x^n + 1$ Then $f(x)$ vanishes for $x = 1$ and $x = T^*(n)$. Since

$$f'(x) = (n+1)x^n - 2nx^{n-1}$$
$$= x^{n-1}[(n+1)x - 2n]$$

is continuous and single-valued for positive x, by Rolle's theorem there is some value x_1 of x in the interval $1 < x < T^*(n)$ for which $f'(x) = 0$. Setting $f'(x) = 0$ and discarding $x = 0$, since it is outside the interval, we find $x_1 = 2n/n+1$. Hence, $T^*(n) > 2n/n+1$ and, for integral n,

$$T(n) > \frac{2n}{n+1} \qquad (4')$$

THEOREM 3. $\lim_{n\to\infty} T(n) = 2 \qquad (5)$

proof: By (4') and (2) we have $2n/n+1 < T(n) < 2$. Since

$$\lim_{n\to\infty} \frac{2n}{n+1} = 2$$

we must have (5).

THEOREM 4. $T(n+1) > T(n) \qquad (5')$

proof: In equation (3) let x be regarded as a variable function $T^*(n)$ of n. Differentiating implicitly with respect to n we obtain

$$\frac{dx}{dn} = \frac{x(2-x)\ln x}{(n+1)x - 2n}$$

Since $1 < \frac{2n}{n+1} < x < 2$ for n greater than unity, we see that $\frac{dx}{dn}$ is positive. Hence, $T^*(n)$ increases monotonically as n increases, and (5') follows.

The following two theorems establish some interesting properties of $T^*(n)$, and also provide less laborious methods for calculating its values.

THEOREM 5. Let n positive numbers $a_0, a_1, \ldots, a_{n-1}$ be chosen arbitrarily, and let succeeding a_i be determined by the recursive relation

$$a_{n+r}^{\,n} = \sum_{i=r}^{n+r-1} a_i^{\,i-r} \qquad (6)$$

Then $\lim_{r\to\infty} a_{n+r} = T(n)$ if the limit exists.

proof: If we write $\lim_{r\to\infty} a_{n+r} = L$, we have also $\lim_{r\to\infty} a_{n+r-i} = L$ for all i. Then (6) gives

$$L^n = \sum_{0}^{n-1} L^i$$

which yields $L = T(n)$.

It will be observed that if we let $n = 2$, $a_0 = a_1 = 1$, we obtain the well-known expression for the golden ratio

$$T(2) = \sqrt{1+\sqrt{1+\sqrt{1+\cdots}}}.$$

THEOREM 6. If a series of numbers a_i is determined by its first n terms and by the recursive relation

$$a_i = \sum_{j=1}^{n} a_{i-j}$$

then the ratio of consecutive terms a_i/a_{i-1} approaches $T(n)$ as i increases indefinitely.

proof: $$\frac{a_i}{a_{i-1}} = \frac{\sum a_{i-j}}{a_{i-1}} = \sum \frac{a_{i-j}}{a_{i-1}}$$

The limit exists since $a_{i-j}/a_{i-1} \le 1$, by which $a_i/a_{i-1} \le n$.

Then, if we write

$$\frac{a_{i-j}}{a_{i-1}} = \frac{a_{i-j}}{a_{i-j+1}} \frac{a_{i-j+1}}{a_{i-j+2}} \cdots \frac{a_{i-2}}{a_{i-1}} = \prod_{k=1}^{j-1} \frac{a_{i-k-1}}{a_{i-k}},$$

we have

$$\frac{a_i}{a_{i-1}} = \sum_{j=1}^{n} \left[\prod_{k=1}^{j-1} \frac{a_{i-k-1}}{a_{i-k}} \right].$$

If we then allow i to increase indefinitely, we have

$$\lim_{i \to \infty} \frac{a_i}{a_{i-1}} = L,$$ and $\lim a_{i-k-1}/a_{i-k}$ is $1/L$. Hence,

$$L = \sum_{j=1}^{n} \left(\prod_{k=1}^{j-1} \frac{1}{L} \right)$$

$$= \sum_{j=1}^{n} \frac{1}{L^{j-1}}$$

Multiplying both sides of the equation by L^{n-1}, we obtain

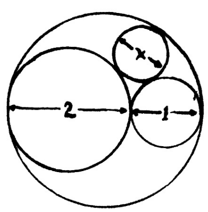
$$L^n = \sum_{j=0}^{n-1} L^j$$

Since L is evidently positive, we have $L = T(n)$.

This theorem parallels the property of the golden ratio of being the limit of the ratio of consecutive terms of the well-known Fibonacci progression 1, 1, 2, 3, 5, 8, 13,... where each term is the sum of the preceeding two terms.

The reader may have observed by now that, although much of the mystery surrounding the golden ratio has arisen from its geometric properties, the above discussion has been limited to analytic considerations. This is because an equation of the third degree is not likely to arise in geometric situations due to the inconstructibility of roots of equations of higher degree than the second. Also, in theorem 5, the existence of the limit still remains to be demonstrated, and the continuity of $T^*(n)$ also requires proof. The serious reader should attempt to fill these gaps and to discover other properties of $T(n)$ by generalizations of properties of the golden ratio.

PROBLEMS -

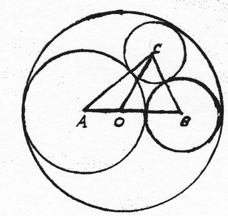


1. The diameters of the circles in the diagram are 1, 2, 3, and x. Find x.

2. Find $\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \ldots = \sum_{1}^{\infty} \frac{1}{n(n+1)}$

3. An urn contains two white balls and three black balls. What is the expectancy of the number of draws required to pick a white ball if -
   (a) balls drawn are not returned to the urn?
   (b) balls drawn are returned to the urn?

SOLUTIONS TO PROBLEMS -



1. $AC = \frac{x+2}{2}$

$AO = \frac{1}{2}$

$OB = 1$

$BC = \frac{x+1}{2}$

$OC = \frac{3-x}{2}$

Applying Stewart's Theorem

$$\frac{3}{2}\left(\frac{3-x}{2}\right)^2 = \left(\frac{x+2}{2}\right)^2 + \frac{1}{2}\left(\frac{x+1}{2}\right)^2 - \frac{3}{4}$$

from which $x = \frac{6}{7}$

* * * * *

2. Since $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$,

we have $$\sum_{1}^{N} \frac{1}{n(n+1)} = \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots + \left(\frac{1}{N} - \frac{1}{N+1}\right)$$

$$= 1 - \frac{1}{N+1}$$

Thus $$\sum_{1}^{\infty} \frac{1}{n(n+1)} = \lim_{N \to \infty} \sum_{1}^{N} \frac{1}{n(n+1)} = 1 - \lim_{N \to \infty} \frac{1}{N+1} = 1.$$

* * * * *

3. $$E_a = \frac{2}{5}(1) + \left(\frac{3}{5}\right)\left(\frac{2}{4}\right)(2) + \left(\frac{3}{5}\right)\left(\frac{2}{4}\right)\left(\frac{2}{3}\right)(3) + \left(\frac{3}{5}\right)\left(\frac{2}{4}\right)\left(\frac{1}{3}\right)(1)(4)$$

$$= 2.$$

$$E_b = (1)\left(\frac{2}{5}\right) + (2)\left(\frac{3}{5}\right)\left(\frac{2}{5}\right) + (3)\left(\frac{3}{5}\right)^2\left(\frac{2}{5}\right) + \ldots$$

$$= \frac{2}{5} \sum_{1}^{\infty} x \left(\frac{3}{5}\right)^{x-1} = \frac{2}{5} \cdot \frac{1}{\left(1-\frac{3}{5}\right)^2} = \frac{5}{2} = 2.5 .$$

* * * * *

In Mathematicks he was greater
Than Tycho Brahe or Erra Pater:
For he, by geometrick scale,
Could take the size of pots of ale;
Resolve, by sines and tangents, straight,
If bread or butter wanted weight;
And wisely tell what hour o' th' day
The clock does strike by algebra.

\- Butler's Hudibras.